

TheraBank Analytics Report

Personal Loan Propensity: Data Methodology & Key Findings

1. Executive Summary & Project Objective

The primary objective of this project is to develop a predictive analytics model for TheraBank to identify liability customers who have a higher propensity to accept a personal loan offer. TheraBank's management wants to explore ways to convert its liability customers to personal loan customers (while retaining them as depositors).

A marketing campaign was previously conducted, and the bank captured data on customer demographics, banking behaviors, and whether they accepted the personal loan. By analyzing this historical data, conducting extensive Exploratory Data Analysis (EDA), applying Weight of Evidence (WoE) and Information Value (IV) transformations, and training a **Naive Bayes Classification model**, the bank can optimize its future marketing campaigns to target the right customers, thereby increasing the conversion rate and reducing marketing costs.

2. Data Preparation & Cleaning Methodology

Before conducting exploratory analysis and modeling, the dataset underwent strict data validation and cleaning protocols to ensure the integrity of the predictive model:

- **Duplicate Detection:** The dataset was scanned for duplicate customer IDs to prevent data leakage and bias. No duplicates were found.
- **Anomaly Correction:** An anomaly was detected in the Experience (in years) variable, where 52 customers had negative experience values (e.g., -1, -2, -3). Since professional experience cannot be negative, these values were treated as entry-level and clipped to 0.
- **Handling Missing Values:** The Family members column contained 18 missing values (NaN). Instead of dropping these records (which could result in information loss), they were isolated into a dedicated categorical class labeled **"Missing"**.

3. Exploratory Data Analysis (EDA) Findings

Bivariate and univariate analysis revealed significant insights regarding customer behaviors and their correlations to loan acceptance:

1. Highly Imbalanced Target Variable: The baseline propensity analysis shows that out of 5,000 customers, only **480 customers (9.60%)** accepted the personal loan, while 4,520 (90.40%) rejected it. This severe class imbalance requires specialized evaluation metrics (like Precision, Recall, and F1-score) rather than just accuracy.

2. Income as a Primary Driver: Analyzing the income distribution reveals a stark contrast between the two groups. Customers who rejected the loan have a median monthly income around \$60K/month. Conversely, customers who accepted the personal loan are predominantly high earners, with a median monthly income of approximately \$140K/month. Income is a strong positive indicator for loan propensity.

3. The Role of Education: Educational background strongly influences conversion rates. Undergraduates (Education Level 1) exhibited the lowest loan acceptance rate. In contrast, Graduates (Level 2) and Professionals (Level 3) showed a markedly higher proportion of loan acceptance, making Education a valuable categorical predictor.

4. Multicollinearity Alert (Age vs. Experience): Pearson correlation analysis flagged a critical multicollinearity issue: Age and Experience possess a near-perfect positive correlation ($r = 0.99$). Having both in a linear or probabilistic model provides redundant information. One of these variables should be dropped during model refinement.

4. Feature Engineering: Weight of Evidence (WoE) & Information Value (IV)

To evaluate the predictive strength of each feature and handle continuous variables robustly, the dataset was subjected to Weight of Evidence (WoE) binning and Information Value (IV) analysis.

$$WOE = \ln(\% \text{ Non-Events} / \% \text{ Events})$$

$$IV = \sum [(\% \text{ Non-Events} - \% \text{ Events}) \times WOE]$$

Features with an IV less than 0.02 are generally considered useless for prediction, while an IV above 0.30 signifies a strong predictor. The table below highlights the computed IV for the TheraBank dataset:

Feature Name	Information Value (IV)	Predictive Power
Income (in K/month)	5.2828	Strong Predictor (Suspiciously High)
CCAvg (Credit Card Avg Spend)	1.6072	Strong Predictor (Suspiciously High)
CD Account	0.6152	Strong Predictor
Education	0.2992	Medium Predictor
Family members	0.0669	Weak Predictor
Mortgage	0.0668	Weak Predictor
Securities Account	0.0051	Useless (Drop)
Age (in years)	0.0048	Useless (Drop)
Experience (in years)	0.0019	Useless (Drop)
Online / CreditCard	< 0.001	Useless (Drop)

Note: Income and CCAvg possess exceptionally high IV scores. This confirms the EDA findings that wealthier customers with higher credit card spending are exponentially more likely to purchase a personal loan from TheraBank. Conversely, demographic features like Age and overall Banking Experience yield almost zero predictive power for this specific product.

5. Predictive Modeling: Naive Bayes Classifier

Following EDA and feature evaluation, a probabilistic **Gaussian Naive Bayes (GaussianNB)** classifier was constructed to automate the identification of prospective loan buyers.

Model Architecture & Pipeline

- **Data Splitting:** The dataset was split into a 70% Training Set and a 30% Testing Set using stratified sampling to preserve the 9.6% acceptance ratio in both subsets.
- **Feature Preprocessing (ColumnTransformer):**
 - *Numerical Features* (Income, CCAvg, Mortgage, Age, Experience) were standardized using `StandardScaler` to ensure normally distributed features with a mean of 0 and variance of 1.

- *Categorical Features* (Education, Family Members, CD Account, etc.) were transformed into binary vectors using `OneHotEncoder`.

- **Classifier:** The `GaussianNB` algorithm was fitted to the preprocessed training data. The model computes the conditional probability of a customer accepting a loan given their specific feature vector, assuming conditional independence between features.

Conclusion & Business Recommendation

By leveraging this machine learning pipeline, TheraBank can transition from mass, untargeted marketing to highly focused, data-driven campaigns. The analysis definitively shows that marketing budgets should prioritize high-income professionals and graduates who actively use credit cards and hold CD accounts. Applying the Naive Bayes model to the current customer base will generate a ranked list of high-probability targets, drastically improving conversion metrics and Return on Investment (ROI).